Experiment 5: Image Segmentation using KMeans (Grayscale and Color)

Aim:

To segment images using KMeans clustering based on grayscale and RGB color information.

Theory:

KMeans clustering is an unsupervised machine learning technique used to group similar data points based on their features. In image segmentation, each pixel is treated as a data point. In grayscale segmentation, each pixel is represented by its intensity value, whereas in color segmentation, each pixel is represented by its RGB color values. The KMeans algorithm partitions pixels into clusters by minimizing the distance between pixel values and cluster centers. This groups visually similar regions together in the image.

Algorithm:

• Start
• Import required libraries (`cv2`, `numpy`, `matplotlib`, `sklearn`)
• Read the image using `cv2.imread()`
• For grayscale segmentation:
• - Convert image to grayscale
• - Reshape into a 1D array of intensity values
• - Apply KMeans clustering and replace pixel values with cluster centers
• For RGB segmentation:
• - Convert image to RGB
• - Reshape into a 2D array of RGB values
• - Apply KMeans clustering and reconstruct segmented image using cluster centers
• Display the original and segmented images
• Plot clustering output (1D for grayscale, 3D for RGB)
• End

Flowchart:

Start → Import Libraries → Read Image → [Grayscale: Convert to Gray → Reshape to 1D] OR [RGB: Convert to RGB → Reshape to 2D] → Apply KMeans → Get Labels and Cluster Centers → Reconstruct Segmented Image → Display Results → End

Viva Questions:

• What features are used in KMeans grayscale vs color segmentation?
• How does KMeans decide the clusters in image segmentation?
• Why do we reshape the image before applying clustering?